

Question number	Answer	Additional guidance	Mark
1(a)(i)	Choose an item. • C is the correct answer A is not the correct answer as acetylcholinesterase is not used in the production of recombinant DNA B is not the correct answer as amylase is not used in the production of recombinant DNA D is not the correct answer as RNA polymerase is not used in the production of recombinant DNA		(1)

Question number	Answer	Additional guidance	Mark
1(a)(ii)	Choose an item. • C is the correct answer A is not the correct answer as DNA is not just found in the nucleus B is not the correct answer as DNA is not just found in the plasmid D is not the correct answer as DNA is not found in the plasmid and nucleus		(1)

Question number	Answer	Additional guidance	Mark
1(a)(iii)	Choose an item. D is the correct answer A is not the correct answer as adjacent nucleotides are not joined by hydrogen bonds B is not the correct answer as adjacent nucleotides are not joined by ionic bonds C A is not the correct answer as adjacent nucleotides are not joined by peptide bonds		(1)

Question number	Answer	Additional guidance	Mark
1(b)	A description that includes three of the following points: - transcription factors are proteins (1) (TF / transcription initiation complex / proteins) bind to {sections of DNA / genes / promotor (region)} (1) {catalysing / causing} {methylation / deacetylation / demethylation / dephosphorylation} of histone (proteins) (1) - preventing the binding of RNA polymerase (to the DNA) (1)	ignore bases / enhancer sequence accept {causing DNA to be wrapped more tightly / heterochromatin} ignore histone modification unqualified accept {stop / decrease} transcription accept makes the DNA inaccessible to RNA polymerase	(3)

